Estimating the volatility σ through an inverse problem for the Black–Scholes equation

Adjoint-state method, numerical solution and optimisation

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Abstract

We aim to identify the volatility σ of the Black–Scholes model from noisy observations of the price of a European call. The problem is written as the minimisation of a least-squares functional, under the constraint of the Black–Scholes PDE. We show that σ is identifiable, then compute the gradient of the functional with the adjoint-state method. The direct problem is solved with finite differences (Crank–Nicolson) and with P_1 finite elements. Both methods are second-order accurate, which we check against the exact solution. Finally, we compare gradient descent with Newton's method, explain why Newton fails far from the solution (non-convexity), and measure how noise affects the estimate.

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1 Model and direct problem

1.1 Context

A European call option gives its holder the right, but not the obligation, to buy an underlying asset at price K (the *strike*) on date T (the *maturity*). If the underlying is worth x at time T, the holder's gain is the *payoff* $(x-K)^+ = \max(x-K, 0)$.

Before maturity, at times $t < T$, the price of this contract depends on a parameter that cannot be observed directly: the volatility σ of the underlying. Option prices, on the other hand, are observed. Recovering σ from these prices is an *inverse problem*: we (partly) know the solution of a PDE and we look for one of its coefficients.

1.2 The Black–Scholes equation

We assume a frictionless market (no transaction costs, perfect liquidity), a constant risk-free rate r, no arbitrage, and an underlying that follows, under the risk-neutral measure $\mathbb{Q}$, a geometric Brownian motion with constant volatility σ:

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t.$$

Let $u(x,t)$ be the price of the call at time t when $S_t = x$. Because there is no arbitrage, the discounted price $e^{-rt}u(S_t,t)$ is a $\mathbb{Q}$-martingale. If $u \in C^{2,1}$, Itô's formula gives

$$\mathrm{d}\big(e^{-rt}u(S_t,t)\big) = e^{-rt}\Big(\partial_t u + rx\,\partial_x u + \tfrac{1}{2}\sigma^2x^2\,\partial_{xx}u - ru\Big)(S_t,t)\,\mathrm{d}t + e^{-rt}\sigma S_t\,\partial_x u(S_t,t)\,\mathrm{d}W_t.$$

An Itô martingale has zero drift, which gives the Black–Scholes PDE

$$\partial_t u + \tfrac{1}{2}\sigma^2x^2\,\partial_{xx}u + rx\,\partial_x u - ru = 0, \qquad x > 0,\ t \in [0,T). \tag{1}$$

1.3 Direct problem on a truncated domain

For the numerical computation, we replace $(0,+\infty)$ with $(0,L)$, where L is large enough. The direct problem is

$$\begin{cases} \partial_t u + \tfrac{1}{2}\sigma^2x^2\,\partial_{xx}u + rx\,\partial_x u - ru = 0, & (x,t) \in (0,L)\times[0,T), \\ u(x,T) = (x-K)^+ =: u_T(x), & x \in [0,L], \\ u(0,t) = 0, & t \in [0,T], \\ u(L,t) = L - Ke^{-r(T-t)} =: u_L(t), & t \in [0,T]. \end{cases} \tag{2}$$

Why these conditions.

- *Terminal condition:* this is the definition of the payoff.
- *Boundary $x = 0$:* a geometric Brownian motion that starts at 0 stays at 0; the option will never be exercised, so it is worth 0.
- *Boundary $x = L$:* put–call parity reads $C(x,t) - P(x,t) = x - Ke^{-r(T-t)}$. For large x, the put P is negligible, so $u(L,t) \approx L - Ke^{-r(T-t)}$.

The data are consistent at the corners of the domain: at $(0,T)$, $u_T(0) = 0$, and at (L,T), $u_T(L) = L - K = u_L(T)$.

Truncation error: Let u_∞ be the solution on $(0,+\infty)$. The difference $w = u_\infty - u$ satisfies (1) on $(0,L)$, with a zero terminal condition, $w(0,t) = 0$ and $w(L,t) = P(L,t)$. The maximum principle for (1) (the term $-ru$ has the right sign) gives

$$\sup_{(0,L)\times[0,T]} |u_\infty - u| \le \sup_{t\in[0,T]} P(L,t).$$

With our parameters ($K = 100$, $L = 4K = 400$, $\sigma = 0.25$, $r = 0.03$, $T = 1$), we find $P(L,0) \approx 6 \cdot 10^{-8}$: the truncation error is negligible compared with all the numerical errors studied below.

1.4 Exact solution and sensitivity to σ

On $(0,+\infty)$, the problem has the explicit Black–Scholes solution. With $\tau = T - t$ and $\mathcal{N}(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-s^2/2}\,\mathrm{d}s$,

$$u_{\mathrm{ex}}(x,t) = x\,\mathcal{N}(d_1) - Ke^{-r\tau}\mathcal{N}(d_2), \qquad d_1 = \frac{\ln(x/K) + (r + \frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}. \tag{3}$$

Its derivative with respect to σ (the *vega*) is

$$\partial_\sigma u_{\mathrm{ex}}(x,t) = x\sqrt{\tau}\,\mathcal{N}'(d_1) > 0 \qquad \text{for } x > 0,\ t < T. \tag{4}$$

It is largest near $x \approx K$ (an "at-the-money" option) and decreases exponentially away from the strike. This property plays a key role in the analysis of the inverse problem.

2 The inverse problem

2.1 Formulation

We have observations u_{obs} on $(0,L)\times(0,T)$ and we look for

$$\sigma^\star \in \arg\min_{\sigma>0} J(\sigma), \qquad J(\sigma) = \frac{1}{2}\int_0^T\int_0^L \big(u_\sigma(x,t) - u_{\mathrm{obs}}(x,t)\big)^2\,\mathrm{d}x\,\mathrm{d}t. \tag{5}$$

Since the map $\sigma \mapsto u_\sigma$ is nonlinear, J is not quadratic in σ and its minimiser has no closed form: we compute it with a descent method, which requires $J'(\sigma)$.

2.2 Identifiability

Proposition 2.1 (Identifiability of σ)**.** *On $(0,+\infty)$, for every $x > 0$ and every $t < T$, the map $\sigma \mapsto u_\sigma(x,t)$ is strictly increasing. In particular, if $\sigma_1 \neq \sigma_2$, then $u_{\sigma_1} \neq u_{\sigma_2}$ on every open subset of $(0,\infty)\times(0,T)$. Hence, for exact data $u_{\mathrm{obs}} = u_{\sigma_{\mathrm{true}}}$, J vanishes at σ_{true} and only there.*

Proof. By (4), the vega is strictly positive: for $\sigma_1 < \sigma_2$, $u_{\sigma_2}(x,t) - u_{\sigma_1}(x,t) = \int_{\sigma_1}^{\sigma_2} \partial_\sigma u_s(x,t)\,\mathrm{d}s > 0$. If $J(\sigma) = 0$, then $u_\sigma = u_{\sigma_{\mathrm{true}}}$ almost everywhere, so $\sigma = \sigma_{\mathrm{true}}$. □

2.3 Synthetic data

Since we have no market data, we generate the observations ourselves. We solve (2) with $\sigma_{\mathrm{true}} = 0.25$ using the scheme of Section 4.2, which gives u^*. We then add multiplicative Gaussian noise at the interior nodes, which makes the experiment more realistic:

$$u_{\mathrm{obs}}(x_i,t^n) = u^*(x_i,t^n)\,\big(1 + a\,\xi_i^n\big), \qquad \xi_i^n \text{ i.i.d.} \sim \mathcal{N}(0,1), \qquad a = 2\,\%. \tag{6}$$

3 Computing the gradient with the adjoint-state method

3.1 Lagrangian

We introduce a multiplier $p(x,t)$ for the PDE, and multipliers $\gamma(x)$, $\lambda(t)$, $\theta(t)$ for the terminal condition and the two boundary conditions:

$$\begin{aligned}\mathcal{L}(u,p,\sigma) = \frac{1}{2}\int_0^T\!\!\int_0^L (u-u_{\text{obs}})^2\,\mathrm{d}x\,\mathrm{d}t + \int_0^T\!\!\int_0^L p\left(\partial_t u + \tfrac{1}{2}\sigma^2x^2\partial_{xx}u + rx\,\partial_x u - ru\right)\mathrm{d}x\,\mathrm{d}t \\ + \int_0^L \gamma(x)\big(u(x,T)-u_T(x)\big)\,\mathrm{d}x + \int_0^T \lambda(t)\,u(0,t)\,\mathrm{d}t + \int_0^T \theta(t)\big(u(L,t)-u_L(t)\big)\,\mathrm{d}t.\end{aligned} \tag{7}$$

If $u = u_\sigma$ solves (2), all the added terms are zero, so

$$\mathcal{L}(u_\sigma, p, \sigma, \gamma, \lambda, \theta) = J(\sigma) \qquad \text{for all } p, \gamma, \lambda, \theta. \tag{8}$$

Differentiating (8) with respect to σ gives

$$J'(\sigma) = \frac{\partial \mathcal{L}}{\partial \sigma} + \frac{\partial \mathcal{L}}{\partial u}\big[\partial_\sigma u_\sigma\big]. \tag{9}$$

The second term involves the unknown sensitivity $\partial_\sigma u_\sigma$. **We choose the multipliers so that $\partial\mathcal{L}/\partial u$ vanishes in every direction**: we are then left with $J'(\sigma) = \partial\mathcal{L}/\partial\sigma$, which is explicit.

3.2 Derivative of the Lagrangian with respect to the state

For a direction φ (a smooth function on $[0,L]\times[0,T]$), we set $\frac{\partial\mathcal{L}}{\partial u}[\varphi] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathcal{L}(u+\varepsilon\varphi,\dots)\big|_{\varepsilon=0}$. Expanding the square in the first term and using the fact that the other terms are linear in u, we get

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial u}[\varphi] = \iint (u-u_{\text{obs}})\varphi + \iint p\left(\partial_t\varphi + \tfrac{1}{2}\sigma^2x^2\partial_{xx}\varphi + rx\,\partial_x\varphi - r\varphi\right) \\ + \int_0^L \gamma\,\varphi(\cdot,T) + \int_0^T \lambda\,\varphi(0,\cdot) + \int_0^T \theta\,\varphi(L,\cdot).\end{aligned} \tag{10}$$

We move all the derivatives from φ onto p by integration by parts, keeping *all* the boundary terms. We write $a(x) = \frac{1}{2}\sigma^2x^2$.

- *Time term (one integration by parts):* $\displaystyle\iint p\,\partial_t\varphi = \int_0^L \big[p\varphi\big]_{t=0}^{t=T}\mathrm{d}x - \iint \partial_t p\,\varphi.$
- *Transport term (one integration by parts):* $\displaystyle\iint rx\,p\,\partial_x\varphi = \int_0^T \big[rx\,p\,\varphi\big]_{x=0}^{x=L}\mathrm{d}t - \iint \partial_x(rxp)\,\varphi.$
- *Diffusion term (two integrations by parts):*

$$\iint a\,p\,\partial_{xx}\varphi = \int_0^T \big[a\,p\,\partial_x\varphi - \partial_x(ap)\,\varphi\big]_{x=0}^{x=L}\mathrm{d}t + \iint \partial_{xx}(ap)\,\varphi.$$

We obtain:

$$\frac{\partial\mathcal{L}}{\partial u}[\varphi] = \iint \Big[(u-u_{\text{obs}}) - \partial_t p + \partial_{xx}(ap) - \partial_x(rxp) - rp\Big]\varphi\,\mathrm{d}x\,\mathrm{d}t \tag{11}$$

$$+ \int_0^L \big(p(x,T)+\gamma(x)\big)\varphi(x,T)\,\mathrm{d}x \;-\; \int_0^L p(x,0)\,\varphi(x,0)\,\mathrm{d}x \tag{12}$$

$$+ \int_0^T \Big[a\,p\,\partial_x\varphi + \big(rxp - \partial_x(ap)\big)\varphi\Big]_{x=0}^{x=L}\mathrm{d}t + \int_0^T \lambda\,\varphi(0,t)\,\mathrm{d}t + \int_0^T \theta\,\varphi(L,t)\,\mathrm{d}t. \tag{13}$$

3.3 The adjoint equation

Interior: Take φ with compact support in $(0,L)\times(0,T)$. The terms (12)–(13) then vanish, and the identity $\iint[\cdots]\varphi = 0$ for every such φ forces the bracket in (11) to be zero. Expanding it gives the **adjoint equation**

$$-\partial_t p + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} p + (2\sigma^2 - r)\,x\,\partial_x p + (\sigma^2 - 2r)\,p = u_{\text{obs}} - u. \tag{14}$$

Boundary conditions. Since the interior bracket is zero, only (12)–(13) remain, and they must vanish for every φ. We collect the coefficient of each trace of φ, using that $a(0)=0$, that $xp=0$ at $x=0$, and that $\partial_x(ap) = \sigma^2 xp + \frac{1}{2}\sigma^2 x^2 \partial_x p = 0$ at $x=0$ for a smooth p.

Trace of φ	**Coefficient**	**Consequence**
$\varphi(x,T)$	$p(x,T)+\gamma(x)$	sets $\gamma = -p(\cdot,T)$
$\varphi(0,t)$	$\lambda(t)$	sets $\lambda = 0$
$\varphi(L,t)$	$rLp(L,t) - \partial_x(ap)(L,t) + \theta(t)$	sets θ
$\varphi(x,0)$	$-p(x,0)$ (no multiplier)	$\boldsymbol{p(x,0)=0}$
$\partial_x\varphi(L,t)$	$\frac{1}{2}\sigma^2 L^2\,p(L,t)$ (no multiplier)	$\boldsymbol{p(L,t)=0}$
$\partial_x\varphi(0,t)$	$\frac{1}{2}\sigma^2\cdot 0^2\cdot p(0,t) = 0$	no condition

On the boundaries where u is prescribed ($t=T$, $x=0$, $x=L$), a multiplier absorbs the corresponding term and places no constraint on p. The conditions on p come from the traces of φ that have no multiplier: the value at $t=0$ (where u is the unknown) and the normal derivative at $x=L$.

These conditions are *necessary*, not only sufficient: the interior of φ, its trace $\varphi(\cdot,0)$ and its derivative $\partial_x\varphi(L,\cdot)$ can be chosen independently of each other. For example, $\varphi(x,t) = g(x)(1 - t/T)$ with $g(0)=g(L)=g'(L)=0$ only activates the term at $t=0$, and $\varphi(x,t) = h(t)\,x(x-L)/L$ with $h(0)=h(T)=0$ only activates the term in $\partial_x\varphi(L,t)$. Each coefficient must therefore vanish on its own.

Remark 3.1 (Boundary $x=0$). No condition on p is imposed at $x=0$. For the numerical scheme, we set $p(0,t)=0$. This is a *choice*, consistent with the right-hand side being zero at $x=0$ (since $u(0,t)=u_{\text{obs}}(0,t)=0$); it is not a consequence of the calculation.

Adjoint system

$$\begin{cases} -\partial_t p + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} p + (2\sigma^2 - r)\,x\,\partial_x p + (\sigma^2 - 2r)\,p = u_{\text{obs}} - u, & (x,t)\in(0,L)\times(0,T], \\ p(x,0) = 0, & x\in(0,L), \\ p(L,t) = 0, \qquad p(0,t) = 0 \ \ \text{(choice)}, & t\in(0,T]. \end{cases} \tag{15}$$

3.4 Gradient formula

Proposition 3.2. *Let u be the solution of (2) and p the solution of (15). Then*

$$\boxed{J'(\sigma) = \sigma \int_0^T \int_0^L x^2\,\partial_{xx} u(x,t)\,p(x,t)\,\mathrm{d}x\,\mathrm{d}t.} \tag{16}$$

Proof using the Lagrangian. With the above choice of p,γ,λ,θ, we have $\partial\mathcal{L}/\partial u = 0$, and (9) gives $J'(\sigma) = \partial\mathcal{L}/\partial\sigma$. In (7), σ appears explicitly only in $\frac{1}{2}\sigma^2 x^2 \partial_{xx} u$, whose derivative is $\sigma x^2 \partial_{xx} u$. □

3.5 Algorithm

Each evaluation of $(J(\sigma), J'(\sigma))$ has three steps:

1. solve the direct problem (2) from $t = T$ back to $t = 0$ to get u;
2. solve the adjoint problem (15) from $t = 0$ to $t = T$, with right-hand side $u_{\text{obs}} - u$, to get p;
3. compute J with (5) and J' with (16).

4 Discretisation

4.1 Mesh and finite differences in space

We set $x_i = i\Delta x$ ($0 \le i \le N_x$, $\Delta x = L/N_x$), $t^n = n\Delta t$ ($0 \le n \le N$, $\Delta t = T/N$) and $u_i^n \approx u(x_i, t^n)$. As a first setting, we take $N_x = N = 100$, so $\Delta x = 4$ and $\Delta t = 0.01$; the strike $K = 100$ is node $i = 25$. Spatial derivatives are approximated by centred differences:

$$\partial_x u(x_i) \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}, \qquad \partial_{xx} u(x_i) \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}.$$

A Taylor expansion shows that both formulas are second-order consistent. The Black–Scholes operator becomes a tridiagonal matrix L_h:

$$(L_h U)_i = \tfrac{1}{2}\sigma^2 x_i^2 \, \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} + r x_i \, \frac{u_{i+1} - u_{i-1}}{2\Delta x} - r u_i.$$

4.2 Crank–Nicolson scheme for the direct problem

Integrating (1) between t^n and t^{n+1} and approximating the integral of the spatial term with the trapezoidal rule, we get

$$\frac{U^{n+1} - U^n}{\Delta t} + \tfrac{1}{2}\big(L_h U^{n+1} + L_h U^n\big) = 0 \quad \Longleftrightarrow \quad \Big(I - \tfrac{\Delta t}{2} L_h\Big) U^n = \Big(I + \tfrac{\Delta t}{2} L_h\Big) U^{n+1}.$$

We start from $U^N = (x_i - K)^+$ and solve a tridiagonal system at each step. The scheme is consistent of order $O(\Delta t^2 + \Delta x^2)$.

4.3 P_1 finite elements

Weak formulation. We multiply (1) by a test function φ that vanishes at $x = 0$ and $x = L$, and integrate over $(0, L)$. Because the coefficient x^2 depends on x, integrating the diffusion term by parts produces *two* terms:

$$\int_0^L \tfrac{1}{2}\sigma^2 x^2 \, \partial_{xx} u \, \varphi \, \mathrm{d}x = -\int_0^L \tfrac{1}{2}\sigma^2 x^2 \, \partial_x u \, \varphi' \, \mathrm{d}x \; - \; \int_0^L \sigma^2 x \, \partial_x u \, \varphi \, \mathrm{d}x.$$

The weak formulation is therefore

$$\int_0^L \partial_t u \, \varphi - \int_0^L \tfrac{1}{2}\sigma^2 x^2 \, \partial_x u \, \varphi' + \int_0^L (r - \sigma^2)\, x \, \partial_x u \, \varphi - \int_0^L r u \, \varphi = 0. \tag{17}$$

Semi-discrete system. With $u = \sum_i u_i(t)\varphi_i$ (the φ_i are hat functions) and the mass, weighted stiffness and convection matrices

$$M_{ji} = \int \varphi_i \varphi_j, \qquad S_{ji} = \int x^2 \varphi_i' \varphi_j', \qquad C_{ji} = \int x \, \varphi_i' \varphi_j,$$

we get $M\dot{U} = K_h U$ with $K_h = \frac{1}{2}\sigma^2 S + (\sigma^2 - r)C + rM$. The integrals in S and C are computed element by element, evaluating x at the midpoint of each element. In time, we use Crank–Nicolson:

$$\Big(M + \tfrac{\Delta t}{2} K_h\Big) U^n = \Big(M - \tfrac{\Delta t}{2} K_h\Big) U^{n+1}.$$

4.4 Discretising the adjoint and the gradient

We solve the adjoint problem (15) with Crank–Nicolson, as for the direct equation. Only the operator, the source term and the direction of time change: we march forward from $t = 0$ to $t = T$. Writing P^n for the vector of the $p_i^n \approx p(x_i, t^n)$:

$$A\,P^{n+1} = B\,P^n + \frac{\Delta t}{2}\big(F^n + F^{n+1}\big), \qquad A = I - \frac{\Delta t}{2} L_h^{\mathrm{adj}}, \qquad B = I + \frac{\Delta t}{2} L_h^{\mathrm{adj}},$$

with $P^0 = 0$, $p_0^n = p_{N_x}^n = 0$, $F^n = U^n - U_{\mathrm{obs}}^n$, and L_h^{adj} the centred discretisation of $\frac{1}{2}\sigma^2 x^2 \partial_{xx} + (2\sigma^2 - r)x\,\partial_x + (\sigma^2 - 2r)$. The source term is $U^n - U_{\mathrm{obs}}^n$ (and not the reverse) because isolating $\partial_t p$ in (14) gives $\partial_t p = \mathcal{L}^{\mathrm{adj}} p - (u_{\mathrm{obs}} - u)$.

The gradient (16) is then approximated by

$$J'(\sigma) \approx \sigma \sum_n \sum_i x_i^2 \, \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}\, p_i^n \, \Delta x \, \Delta t.$$

5 Validation of the direct problem

5.1 Solution and price at the strike

Figure 1 shows the Crank–Nicolson solution at three times. At $t = T$, it is the payoff; at earlier times, diffusion smooths out the kink, and the curve lies strictly above the payoff around K: this is the *time value* of the option, which comes from uncertainty. At $x = L$, we recover the boundary condition, $u(L, 0) = 400 - 100e^{-0.03} \approx 302.96$.

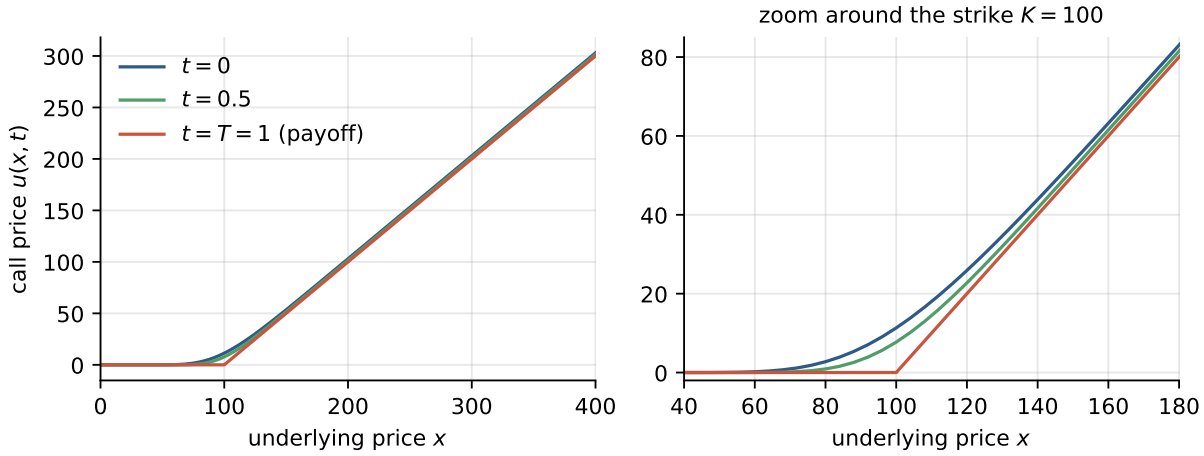


Figure 1: Solution of the direct problem (Crank–Nicolson, $N_x = N = 100$) at $t = 0$, $t = 0.5$ and $t = T$. Right: zoom around the strike.

	Crank–Nicolson	P_1 finite elements	Exact (3)
$u(K, 0)$	11.3174	11.3368	11.3485
Error at $x = K$	$3.1 \cdot 10^{-2}$	$1.2 \cdot 10^{-2}$	–
Maximum error at $t = 0$	$3.1 \cdot 10^{-2}$ (at $x = 96$)	$2.4 \cdot 10^{-2}$ (at $x = 76$)	–

Table 1: Price today of the at-the-money call, for $N_x = N = 100$.

5.2 Convergence study

We refine space and time together ($N = N_x$, so $\Delta t = \Delta x/400$) and measure the error at $t = 0$ against the exact solution, in the discrete L^2 norm $\left(\Delta x \sum_i e_i^2\right)^{1/2}$ and in the L^∞ norm.

Both methods are **asymptotically second-order** in both the L^2 and the L^∞ norms: a regression on the four finest grids gives 2.00 in all four cases (Table 2, Figure 2).

	Crank–Nicolson				P_1 finite elements			
Δx	L^2	order	L^∞	order	L^2	order	L^∞	order
16	1.00	–	$1.74 \cdot 10^{-1}$	–	1.65	–	$2.66 \cdot 10^{-1}$	–
8	$2.45 \cdot 10^{-1}$	2.03	$3.13 \cdot 10^{-2}$	2.47	$5.16 \cdot 10^{-1}$	1.68	$8.10 \cdot 10^{-2}$	1.72
4	$1.80 \cdot 10^{-1}$	0.44	$3.12 \cdot 10^{-2}$	0.01	$1.42 \cdot 10^{-1}$	1.86	$2.40 \cdot 10^{-2}$	1.76
2	$4.50 \cdot 10^{-2}$	2.00	$7.80 \cdot 10^{-3}$	2.00	$3.55 \cdot 10^{-2}$	2.00	$6.02 \cdot 10^{-3}$	1.99
1	$1.12 \cdot 10^{-2}$	2.00	$1.95 \cdot 10^{-3}$	2.00	$8.88 \cdot 10^{-3}$	2.00	$1.50 \cdot 10^{-3}$	2.00
0.5	$2.81 \cdot 10^{-3}$	2.00	$4.87 \cdot 10^{-4}$	2.00	$2.22 \cdot 10^{-3}$	2.00	$3.76 \cdot 10^{-4}$	2.00

Table 2: Errors at $t = 0$, with $\Delta t = \Delta x/400$. The order in row Δx is $\log_2\left(e_{2\Delta x}/e_{\Delta x}\right)$.

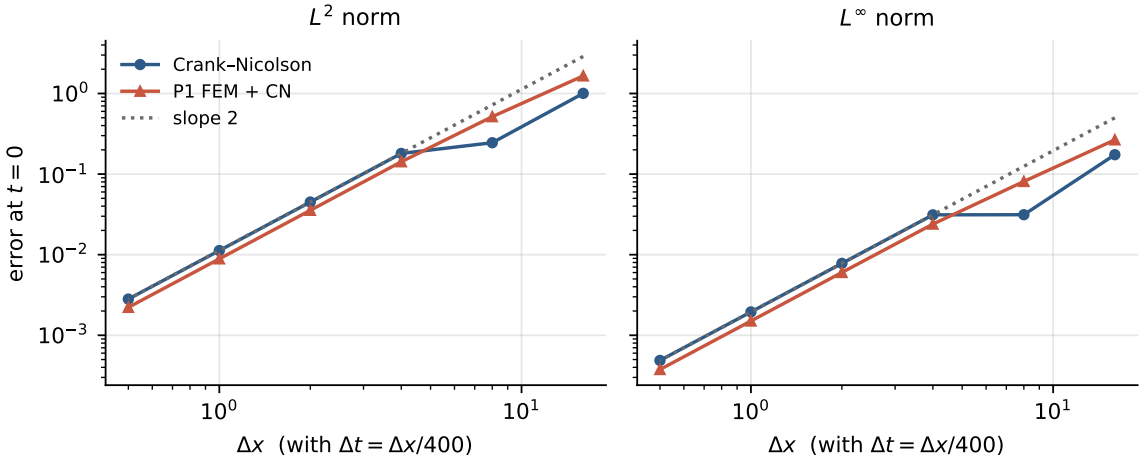


Figure 2: Convergence at $t = 0$ in the L^2 and L^∞ norms. The dotted line has slope 2.

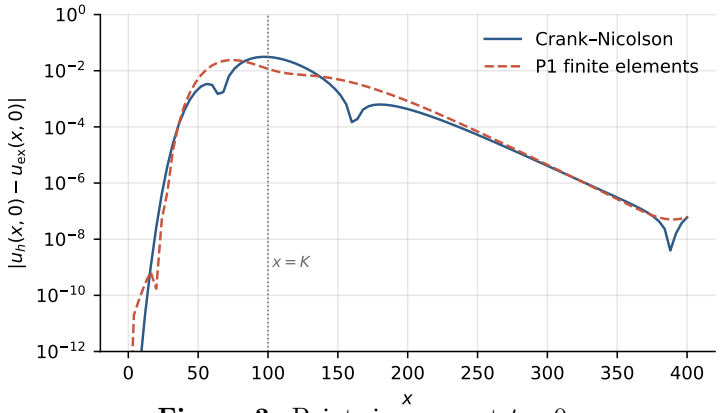


Figure 3: Pointwise error at $t = 0$.

6 Optimisation algorithms

We minimise J with two methods: gradient descent and Newton–Raphson. The observations are generated as in (6), with $a = 2\,\%$. Each evaluation of J' costs one direct solve (from T to 0) and one adjoint solve (from 0 to T).

6.1 Gradient descent

At each iteration k:

1. solve the direct PDE with σ_k to get $u^{(k)}$;
2. solve the adjoint PDE with σ_k and right-hand side $u_{\text{obs}} - u^{(k)}$ to get $p^{(k)}$;
3. compute the gradient $g^{(k)} = \sigma_k \iint x^2\, \partial_{xx} u^{(k)}\, p^{(k)}$;
4. if $|g^{(k)}| < 10^{-5}$, stop; otherwise, update $\sigma_{k+1} = \sigma_k - \eta\, g^{(k)}$, with $\eta = 10^{-5}$.

σ_0	$\sigma^\star$	iterations
0.05	0.25055	69
0.20	0.25055	60
0.85	0.25055	62

Table 3: Gradient descent for three initial values.

Comments:

- **Same result whatever σ_0.** The algorithm stops where $J' = 0$, which depends only on the data. We find $\sigma^\star = 0.25055$ rather than 0.25 because of the noise (Section 6.3).
- **Linear convergence.** Near the minimum, the distance is multiplied at each iteration by $1 - \eta J''(\sigma^\star) \approx 1 - 10^{-5} \times 2.77 \cdot 10^4 \approx 0.72$. This gives the straight lines in Figure 4.

6.2 Newton–Raphson algorithm

We replace the update with

$$\sigma_{k+1} = \sigma_k - \frac{J'(\sigma_k)}{J''(\sigma_k)}, \qquad J''(\sigma) \approx \frac{J(\sigma + h) - 2J(\sigma) + J(\sigma - h)}{h^2}, \quad h = 10^{-4},$$

together with a step-size search: we halve the step as long as J does not decrease.

Starting from $\sigma_0 = 0.9$. We obtain $\sigma^\star = 0.250547$ in 7 iterations, against 62 for gradient descent. This is *the same* point as with gradient descent: Newton is faster, but not less accurate.

Starting from $\sigma_0 = 0.08$. The basic version stops immediately:

```
0  sigma=0.080000  J=2.0766e+03  grad=-1.8303e+03
 step too small, stop.
```

Explanation. J is not convex for $\sigma < 0.11$ (Figure 5): at 0.08, we have $J' < 0$ (so σ should increase), but also $J'' \approx -9.3 \cdot 10^3 < 0$. The step $-J'/J''$ is therefore *negative* and makes J increase, so the step-size search fails. The problem comes from the non-convexity of J, not from the choice of h.

Fix. If $J'' \leq 0$, we replace the Newton step with a small descent step, $-\operatorname{sign}(J') \cdot 0.1\, \sigma_k$. The algorithm then leaves the non-convex region in 4 steps and converges normally:

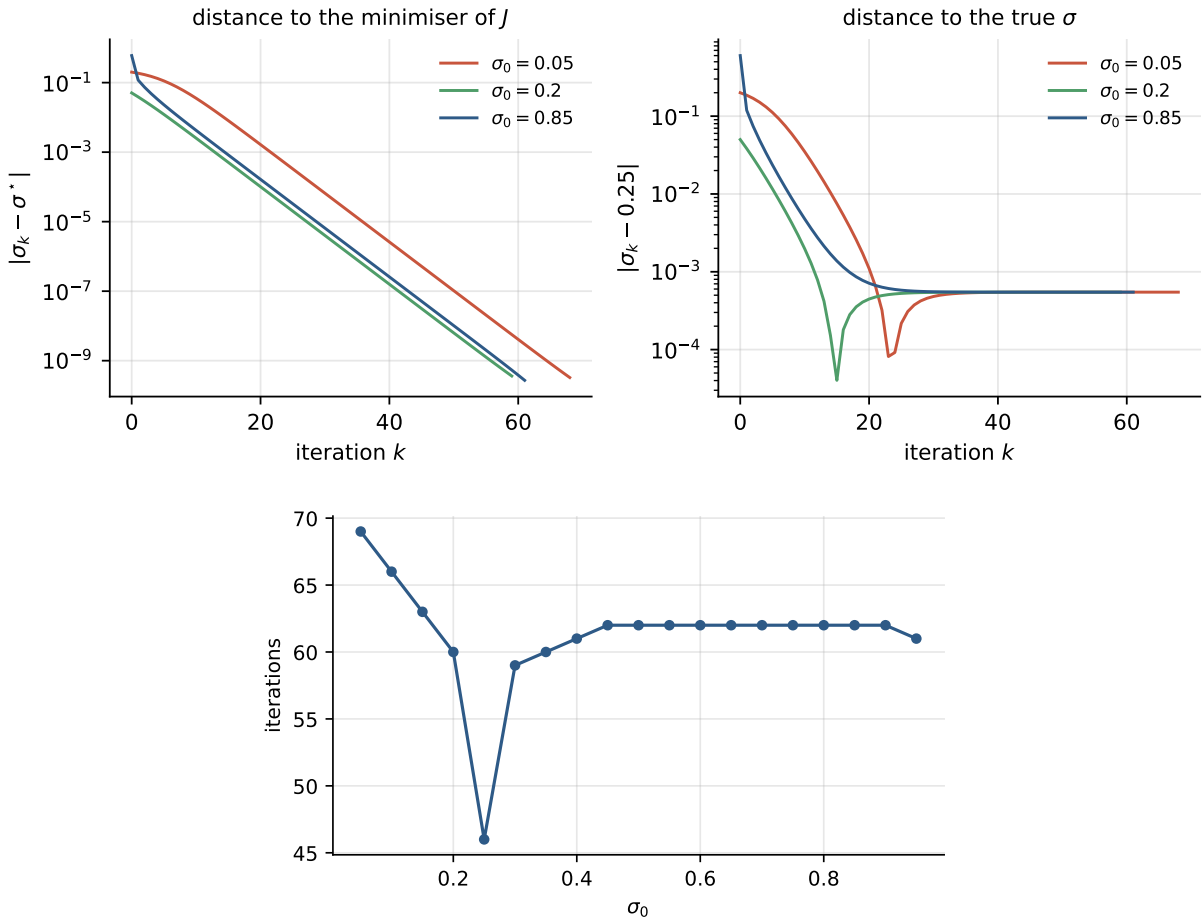


Figure 4: Top left: distance to the minimiser over the iterations. Top right: distance to the true value 0.25; for $\sigma_0 < 0.25$, σ_k overshoots 0.25 (the dip in the curve) and then settles at $\sigma^\star = 0.25055$. Bottom: number of iterations as a function of σ_0.

```
 0  sigma=0.080000  J=2.0766e+03  grad=-1.8303e+03
 4  sigma=0.117128  J=2.0044e+03  grad=-1.9853e+03
 5  sigma=0.311556  J=1.8898e+03  grad=2.0068e+03
 7  sigma=0.250802  J=1.8316e+03  grad=7.0263e+00
10  sigma=0.250547  J=1.8316e+03  grad=4.9018e-07
```

6.3 Effect of noise

To check whether the gap $\sigma^\star - 0.25$ is a flaw of the method, we repeat the estimation with 20 different noise samples.

noise a	mean of $\sigma^\star$	standard deviation
0	0.25000	–
1 %	0.24997	$1.9 \cdot 10^{-4}$
2 %	0.24994	$3.8 \cdot 10^{-4}$
5 %	0.24984	$9.5 \cdot 10^{-4}$
10 %	0.24967	$1.9 \cdot 10^{-3}$

Table 4: Estimates of σ over 20 noise samples ($\sigma_{\text{true}} = 0.25$).

- Without noise, we recover exactly 0.25 (Proposition 2.1).

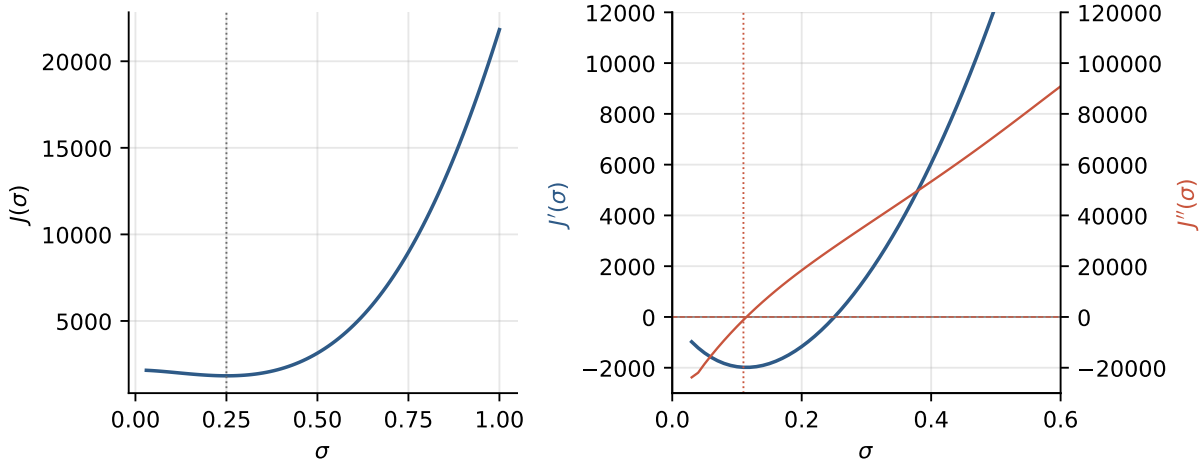


Figure 5: Left: $J(\sigma)$. Right: J' (blue) and J'' (red). $J'' < 0$ for $\sigma < 0.11$.

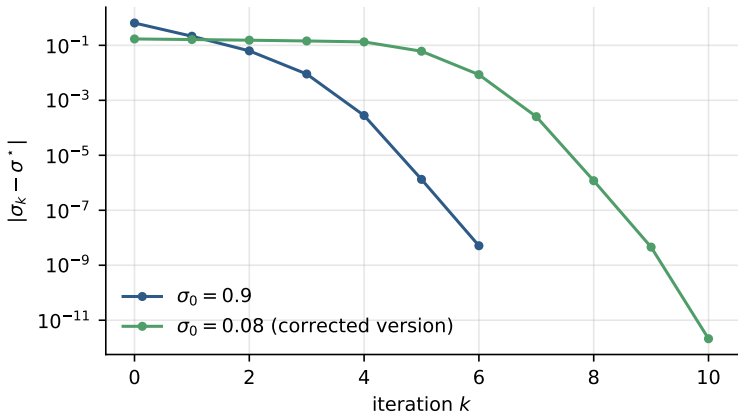


Figure 6: Newton–Raphson: distance to the minimiser for $\sigma_0 = 0.9$ and $\sigma_0 = 0.08$ (corrected version).

- The means stay close to 0.25: the method shows no significant bias.
- The standard deviation is proportional to the noise level (about $0.02\,a$).
- The value 0.25055 found above is therefore just a fluctuation caused by the noise.

7 Conclusion

We estimated the volatility σ by minimising the gap between computed and observed prices. The gradient is obtained with the adjoint-state method: one direct solve and one adjoint solve are enough, with $J'(\sigma) = \sigma \iint x^2\, \partial_{xx} u\, p$.

Finite elements (with a complete weak formulation) and Crank–Nicolson are both second-order accurate. Gradient descent and Newton converge to the same $\sigma^\star$; Newton needs about ten times fewer iterations, but it must be protected against the non-convexity of J for $\sigma < 0.11$. The error on σ is proportional to the noise in the data.

A natural next step would be to use real market data (today's prices for several strikes and maturities) and to look for a local volatility $\sigma(x, t)$. The problem then becomes ill-posed and needs regularisation, but the adjoint method still gives the full gradient at the same cost.